# Research Statement

After becoming fascinated with physics at Allameh Helli High School, I opted to pursue it further. In 2018, winning a gold medal in the National Physics Olympiad exempted me from the national university entrance exam, allowing me to study physics at the Young Students' Club for an additional year. During this time, I explored various topics such as optics, special relativity, thermodynamics, and electromagnetism. Due to my interest in engineering and the wide range of fields of study that electrical engineering offers, I chose electrical engineering at the university. As I studied various fields within electrical engineering, my focus shifted toward machine learning, signal processing, and computer vision.

I completed various courses, including convex optimization, computer vision, deep learning, and machine learning. In these courses, I became acquainted with some of the most widely used image and signal processing methods, theories in machine learning, and challenges faced in big data and real-time decision-making.

I initiated my undergraduate project under the kind supervision of Dr. Hashemi, which involved carefully inspecting commonly used items for medical plating purposes, such as screws and bolts, using computer vision and image processing.

I completed my MSc thesis at Sharif University of Technology under the kind supervision of Dr. Fotowat-Ahmady and Dr. Khalaj. In that project, I created an environment to test and develop new autonomous driving algorithms and design modules used for Advanced Driving Assistance Systems (ADAS).

One of the main challenges in autonomous driving is developing fast and reliable sensor-fusion and navigation algorithms that can operate in unknown environments while addressing issues of robustness, uncertainty, and safe decision-making.

These experiences motivated my interest in broader questions concerning the reliability, behavior, and safety of intelligent agents. I am particularly interested in your work on trustworthy AI agents, multi-agent cooperation, and the safety challenges arising from increasingly autonomous AI systems.

I believe my background in autonomous systems, machine learning, and sensor fusion would allow me to contribute effectively to the research group while continuing to develop as a researcher. I am highly motivated to pursue doctoral research and would welcome the opportunity to contribute to COMPASS Research Group.

Thank you for your attention.

Dariush Ghaemi